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Unlocking CO_2 Injection in Depleted Reservoirs: A Flow Assurance Modeling Approach for Carbon Capture and Storage

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Abstract

Carbon, Capture, and Storage (CCS) plays a pivotal role in unlocking production from sour gas fields. For a recent PTTEP greenfield development in Sarawak, Malaysia, hydrocarbon gas containing carbon dioxide (CO_2) above 15 mol% must be treated to meet sale specifications. To develop the hydrocarbon resources with the least Greenhouse Gas (GHG) intensity, CO_2 captured from gas production will need to be transported and injected offshore into depleted reservoirs.

The behavior of CO_2 up to the storage reservoirs must be extensively investigated to overcome challenges on corrosion, injectivity, and low temperature. A PVT analysis was originally done in the front-end design stage to define the phase envelope of the injected CO_2 in relation to the contaminants from the gas treatment system. Subsequently, steady state and dynamic flow assurance assessments of the integrated CO_2 transport and injection network were conducted to determine the safe operating window for the overall system. Key findings were then translated into functional design requirements for the CCS facilities such as the transport pipeline and injection platform.

To mitigate hydrate risks at the well bottom-hole and keep CO_2 along the transport pipeline and injection wellhead platform in supercritical phase, CO_2 injection must closely adhere to the established pressure, temperature, and flow rate envelopes based on the flow assurance analysis. Once all storage reservoirs are sufficiently pressurized, the arrival temperature of CO_2 at the injection platform no longer needs to be controlled which will allow CO_2 to be injected at seabed temperature with no adverse risks in the wellbore.

Several facility design options to achieve the desired CO_2 injection pressure and temperature at the remote wellhead platform were then assessed resulting in the most cost-effective and energy-efficient decision to insulate the subsea transport pipeline. Dynamic flow assurance study results were also included to operating philosophies enabling the CO_2 transport system to be started up and restarted even under the most onerous pipeline shut-in scenarios and environmental circumstances. For possible issues with trace impurities in the captured CO_2, such as TEG, the fluid specifications at the inlet of the transport pipeline are tightly controlled and considered satisfactory in minimizing the severe corrosion risk in the downstream facilities.

In developing this greenfield initiative with minimal GHG emission, understanding the flow behaviors along the transport and injection network is crucial to enable up to 2.0 MTPA of CO_2 injection into depleted

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hydrocarbon reservoirs. The lessons learned from this project can be used as a starting framework for future developments to overcome similar obstacles associated with CO_2 injection.

Introduction

As the demand for Carbon, Capture, and Storage (CCS) continues to grow to abate global carbon dioxide (CO_2) emissions, more depleted hydrocarbon reservoirs are being converted into CO_2 storage reservoirs. Unlike saline aquifers, depleted reservoirs can be rapidly characterized and converted to CO_2 storages due to the vast amount of geological data accumulated over the production phase. Despite their developmental benefits, depleted reservoirs are often low in pressure at the start of CO_2 injection. For a recent PTTEP greenfield development in Sarawak, Malaysia, hydrocarbon gas containing carbon dioxide (CO_2) above 15 mol% must be treated to meet sale specifications. In order to develop the hydrocarbon resources with the least Greenhouse Gas (GHG) intensity, CO_2 must be captured and transported 40 km offshore for injection into depleted hydrocarbon reservoirs. Through a flow assurance modelling perspective, this paper aims to identify the thermohydraulic challenges associated with CO_2 injection in depleted storage sites and to provide a conceptual design framework for establishing a safe operating window to overcome challenges with low temperature, injectivity, and corrosion in low-pressure reservoirs during steady state and transient scenarios.

Project Overview

The CO_2 that will be sequestered as part of this greenfield initiative will be captured by a solvent-based gas treatment unit at an onshore processing plant. Once captured, CO_2 is then compressed and dehydrated before flowing into the transport pipeline. Although CO_2 is transported to the injection platform in dense / supercritical phase, a stringent water content is still required to prevent free water drop out and hydrate formation along the pipeline and offshore facility during low temperature scenarios (e.g., pressure letdowns, depressurization, etc.).

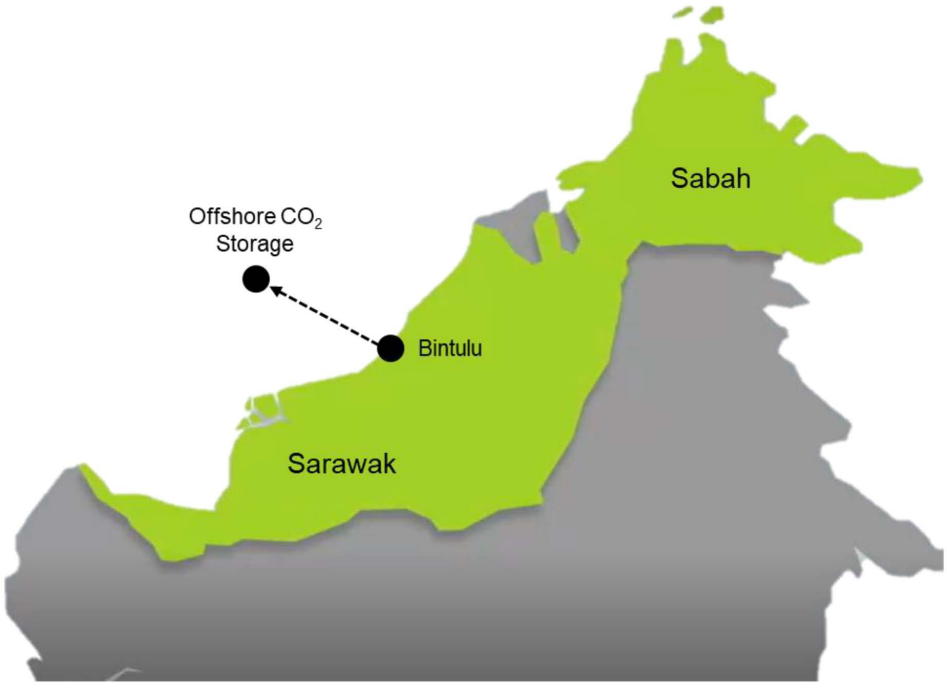


Figure 1—Project Location and Offshore CO_2 Storage Site

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The CO_2 storage site for this development is located 40 km offshore from the gas processing plant and consists of multiple sandstone hydrocarbon reservoirs that will be depleted at the start of CO_2 injection. Although the reservoirs have been producing for more than fifteen (15) years, the integrities of the caprock and fault at this site are still effective. At depletion, the reservoirs provide sufficient storage capacities for sequestrating CO_2 from the gas processing plant throughout its design life. All existing hydrocarbon production wells at the offshore CO_2 storage site will be plugged and abandoned prior to starting CO_2 injection to minimize potential leak paths. New injection wells designed specifically for CO_2 service will be drilled and completed for each reservoir.

For this specific CO_2 storage site, multiple depleted reservoirs will be utilized. The reservoirs vary in depth, pressure, and permeability, but they can be categorized into three (3) groups: i) *shallow* reservoirs, ii) *intermediate* reservoirs, and iii) *deep* reservoirs. The representative features of each category of CO_2 storage reservoir are summarized in Table 1.

Table 1—Main Properties of the Shallow, Intermediate, and Deep Reservoirs.

Reservoir	Depth / TVD (m)	Start of Inj. Pressure (barg)	End of Inj. Pressure (barg)
Shallow	~650	~25	~70
Intermediate	~1550	~30	~170
Deep	~1700	~50	~190

All reservoir targets are planned to be injected simultaneously to reduce the risk of leakage from differential pressure. The CO_2 flow into each reservoir will be closely regulated based on the size of each storage compartment to ensure that the pressures of all compartments build up uniformly.

CO_2 Fluid Behavior Characterization

Understanding the CO_2 fluid characteristics in the transport pipeline and the wellbore is vital for establishing a safe operating window for the CO_2 injection network. Before entering the transport pipeline to the injection site, the CO_2 purity will be above 99 mol% (with < 1 mol% hydrocarbon carryover) from the CO_2 capture and dehydration technologies. The key contaminants with the CO_2 fluid stream exported from the onshore gas processing plant are shown in Table 2.

Table 2—Key Contaminants in the CO_2 Stream Entering the Transport Pipeline.

Species	Periodicity
H_2O	Continuous
H_2S	Continuous
TEG	Continuous

Using the above components, the PVT behavior of the transported CO_2 fluid stream was investigated with Multiflash. Figure 2 depicts the effects of H_2O, H_2S, and TEG on the specific CO_2 Pressure-Temperature (P-T) phase envelope using the Peng-Robinson equation of state. From the P-T phase diagram, the dewpoint curve for the CO_2 transported and injected in this development is observed to be wider than that for pure CO_2. Due to the trace amount of heavy hydrocarbons (C6+) in the fluid stream, a liquid CO_2-hydrocarbon mixture can form at low pressures and temperatures below 30 barg and 0°C respectively. More significantly, the presence of TEG in the CO_2 fluid stream can also promote the formation of a water-glycol mixture even at pressures above the critical point of CO_2 (73 barg). As a result, the minimum operating pressure of the

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CO_2 pipeline and injection platform was considered above 80 barg to ensure that CO_2 can be transported in dense / supercritical phase without the risk for two-phase flow and corrosion.

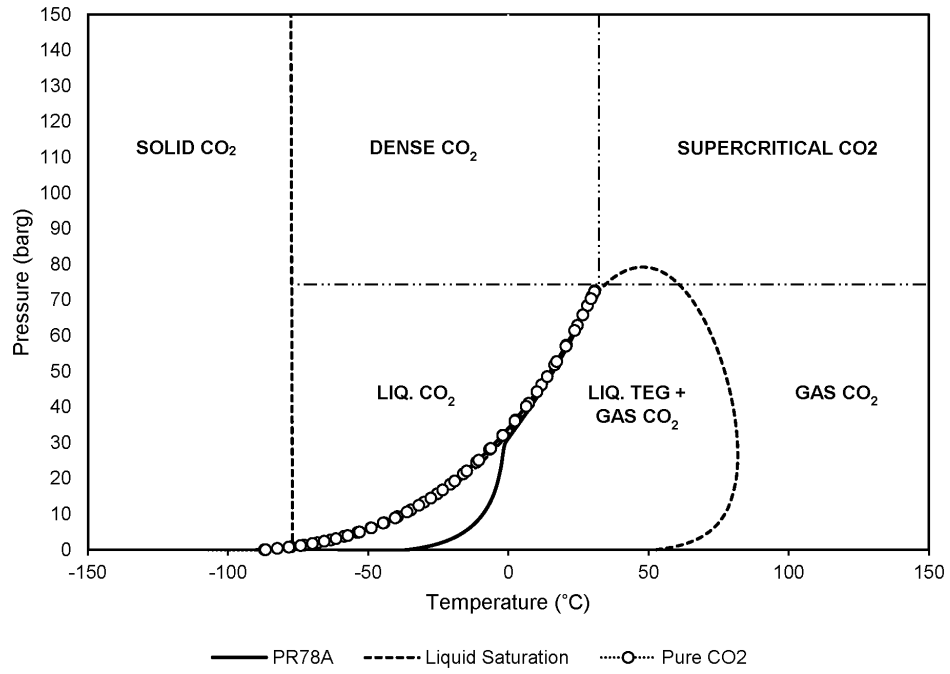


Figure 2—CO_2 Pressure-Temperature (P-T) Phase Diagram

Although CO_2 is maintained in single phase during normal operations, a small amount of the water-glycol mixture (approx. 0.001 vol.%) may form in the pipeline. With the stringent water content specification at the inlet of the transport pipeline (40 ppmv), the mass fraction of TEG in the condensed aqueous phase will exceed 85 wt.% and reduce the potential for CO_2 corrosion by over 85%.[1,2] Table 3 presents the estimated corrosivity of CO_2 in the presence of the TEG mixture for both pipeline shut-in and upset conditions.

Table 3—CO_2 Corrosivity for Pipeline Shut-In and Upset Conditions

Pipeline Conditions	Pipeline Shut-In		Pipeline Upset	
	No TEG	W/. TEG	No. TEG	W/. TEG
CO_2 Corrosivity	Baseline	+0.00%	+220.92%	+221.02%

Notes:

I) *Shut-In Conditions: Minimum operating pressure & temperature*

II) *Upset Conditions: Maximum operating pressure & temperature and five (5) times the normal water content*

The table above demonstrates that the water content in the pipeline is the main driver for CO_2 corrosion. Moreover, the presence of TEG has no substantial impact on the corrosivity of CO_2. For this project, additional material testing conducted on the line pipe also indicates that the corrosion rate at the pipeline upset conditions is less than 0.013 mm/y. Thus, the components formed by the reaction between TEG and water is not corrosive beyond the design allowances of the CO_2 injection network.

Integrated CO_2 Flow Assurance Analysis

After characterizing the fluid, a flow assurance analysis was performed using OLGA to identify the thermohydraulic behavior of CO_2 in the subsea transport pipeline, each injection well (along with its

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surrounding formation), and its near-wellbore region. A single, integrated pipeline-well OLGA model was used to simulate all components of the CO_2 injection network from the discharge of the onshore CO_2 compressors to each reservoir contact. Critical to the analysis, the CO_2 injection choke valves were also included into the integrated model to capture the Joule-Thompson (J-T) cooling effects on the fluid temperature when CO_2 is expanded across the choke valve from the pipeline operating pressure to the pressure at each wellhead. The integrated model's boundary conditions were based on the reservoir parameters shown in Table 1.

A key major outcome of the integrated flow assurance analysis is to define a safe operating envelope (flow rate, pressure, and temperature) for the CO_2 transport and injection network of this development. In particular, the network's operating conditions (especially upstream of the choke valves at the injection platform) will be established to ensure the CO_2 fluid temperature remains above the following design constraints throughout the development's life cycle (i.e., early-life and late-life operation):

I. Minimum design temperature of -46°C for piping after the choke valve connected to injection well
II. Minimum design temperature of -7°C for typical Surface Controlled Subsurface Safety Valve (SCSSV)

In addition, hydrate formation risks were also assessed at critical points in the network including downstream of each CO_2 injection choke valve and at the bottom-hole of each well. Figure 3 presents the CO_2 hydrate curves considered in this development. At a water content of ≤ 40 ppmv, CO_2 hydrates downstream of the injection choke valves are highly unlikely to form as the conditions at the wellheads, estimated from the starting reservoir pressures (Table 1), are outside of the hydrate formation zone (P < 15 barg and T < -35°C). However, there is a possibility that the injected CO_2 will encounter formation / connate water trapped near or around the perforation zone at the well bottom-hole which will re-saturate the injected CO_2 stream. As a conservative approach, the CO_2 fluid temperature at the well bottom-hole is also limited above 10°C to minimize hydrate risk near the reservoir contact.

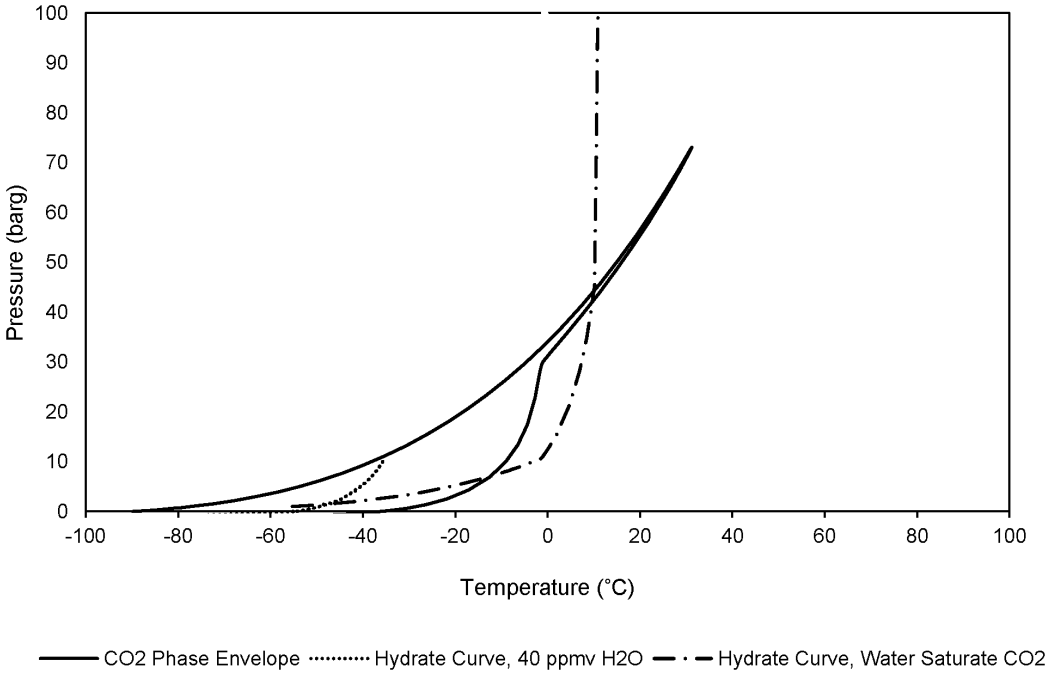


Figure 3—CO_2 Hydrate Formation Curves

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CO_2 Thermohydraulic Behavior in Depleted Reservoirs

From the design constraints in the previous section, a steady state analysis was first conducted to investigate the option of injecting CO_2 at seabed temperature into the storage reservoirs. With the pressure upstream of the choke valve on the injection platform set at 80 barg, the early-life tubing pressure and temperature profiles for CO_2 injection at various flow rates between the turndown and design capacities of each well into the *intermediate* reservoir (P_{res} ~30 barg) and *deep* reservoir (P_{res} ~50 barg) are shown in Figure 4.

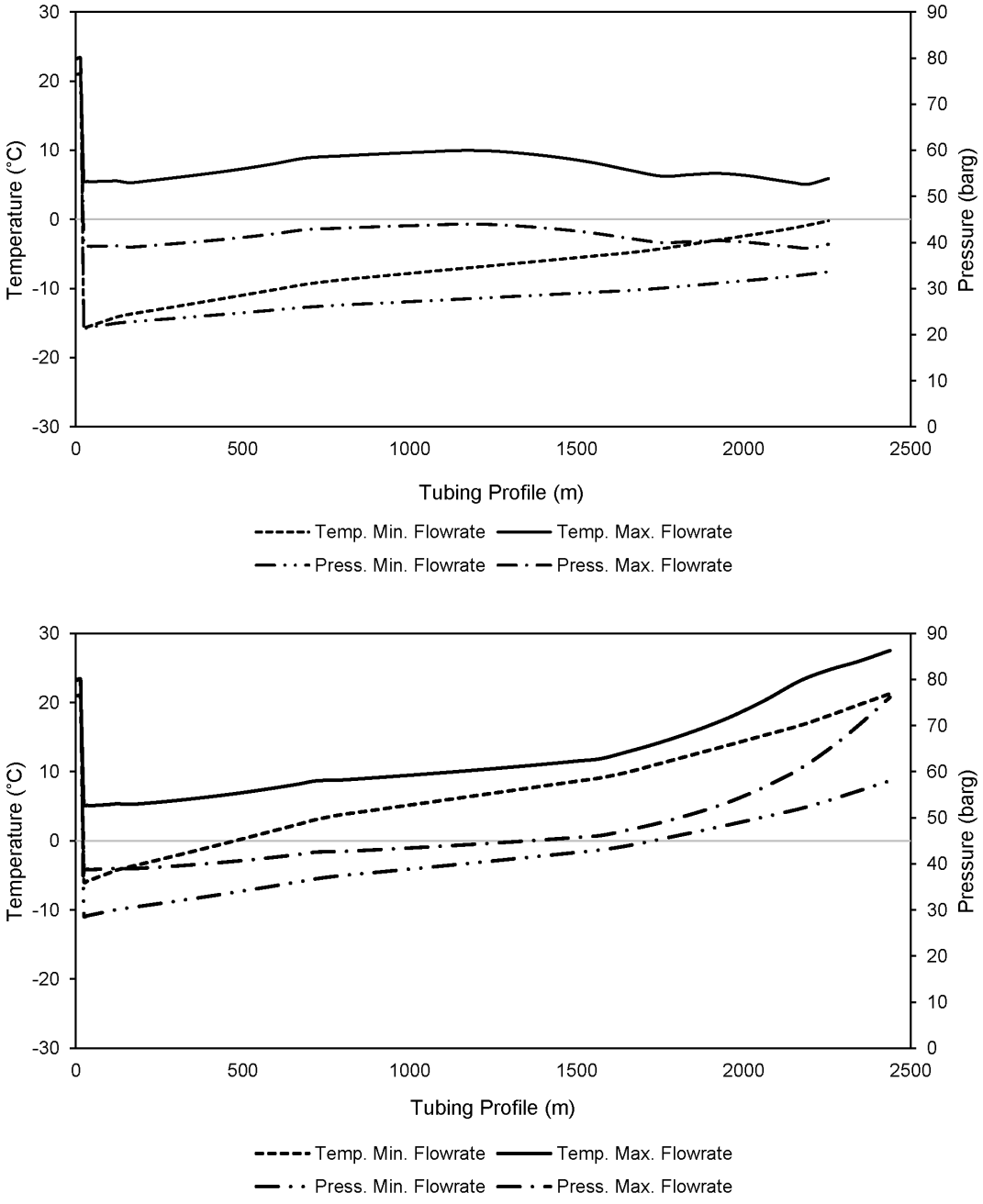


Figure 4—Tubing P-T Profiles for Early-Life CO_2 Injection into the *Intermediate* Reservoir (Top) and *Deep* Reservoir (Bottom)

The Pressure-Temperature (P-T) profiles in Figure 4 demonstrate that the CO_2 fluid temperatures will remain above the minimum design temperature of the piping downstream of the choke valves (-46°C) if it were injected at seabed temperature into the *intermediate* and *deep* reservoirs. However, low CO_2 fluid temperatures are expected at both the SCSSV (< -7°C) and well bottom-hole (< 10°C) for the *intermediate*

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reservoir. The CO_2 fluid temperature at the SCSSV is dictated by the J-T effect across the choke valve as the pressure is reduced from 80 barg to the early-life Injection Tubing Head Pressure (ITHP) at the wellhead. At the turndown injection capacity, minor contributions from the friction loss in the injection tubing and pressure drop across the perforation at the reservoir contact will lead to a relatively low ITHP and subsequently a CO_2 fluid temperature at the SCSSV as low as -12.7°C. At higher flow rates up to the well's design capacity, a higher friction loss in the tubing will lead to a higher ITHP at the wellhead which will reduce the J-T effect across the choke valve resulting in temperatures above -7°C at the SCSSV. Evident from the behavior of CO_2 in depleted reservoirs, the CO_2 fluid temperature above the SCSSV can be controlled above the minimum design temperature constraint by one of the following:

- Reducing the J-T cooling effect across the injection choke valve; or
- Increasing the turndown flow rate of each well during early-life operation; or
- Setting the SCSSV deeper in the injection tubing to allow the CO_2 fluid temperature to increase prior to contacting the valve.

Regardless of the temperature at the SCSSV, the temperature at the well bottom-hole for the *intermediate* reservoir will always be below 10°C. Even though the CO_2 fluid temperature at the turndown flow rate is observed to rise along the injection tubing, the J-T cooling effect at the wellhead cannot be offset by the temperature increment due to the compression effect from gravity and heat transfer into the wellbore from the surrounding rock formation. On the contrary, the CO_2 fluid temperature will continue to drop as the gas is continuously expanded along the wellbore for CO_2 injection at higher flow rates. Under this scenario, heat transfer from the surrounding is also less effective resulting in a well bottom-hole temperature lower than 10°C.

Conversely, CO_2 fluid temperatures lower than the design limits at both the SCSSV and well bottom-hole are not observed in the *deep* reservoir. As the CO_2 density increases with the reservoir pressure, the pressure of the CO_2 stream will continue to rise throughout the trajectory as the compression effect from gravity outweighs the resistance from the friction loss in the injection tubing. In this situation, the CO_2 fluid temperature will continue to increase as it flows through the wellbore.

In summary, CO_2 injection at seabed temperature is not feasible for all reservoir conditions except the *deep* reservoir in this development. As a result, the temperature of the CO_2 fluid stream arriving at the injection platform must be increased to compensate the J-T effect across the choke valve and subsequent cooling in the wellbore from friction loss in the injection tubing.

Early-Life Operation

To define the early-life operating conditions, a sensitivity analysis investigating the effects of the CO_2 injection temperature (upstream of choke valve) on the well bottom-hole temperature was conducted, and the results are shown in Figure 5. For this study, the pressure upstream of the choke valve was controlled at 80 barg while the properties of the *intermediate* reservoir (start of injection reservoir pressure) were used as the boundary conditions due to it being the main injection target with the largest storage volume. Figure 5 shows that 50°C is the most optimal CO_2 injection temperature upstream of the choke valve with respect to the *intermediate* reservoir. At this injection temperature, the well bottom-hole temperature is expected to be 17°C. Increasing the CO_2 injection temperature to 60°C can increase the difference between the bottom-hole temperature and the design constraint, but it will also decrease the maximum flow rate achievable in each injection well (~20% reduction) – due to the increased CO_2 volumetric flow rate and friction loss in the injection tubing – which is undesirable.

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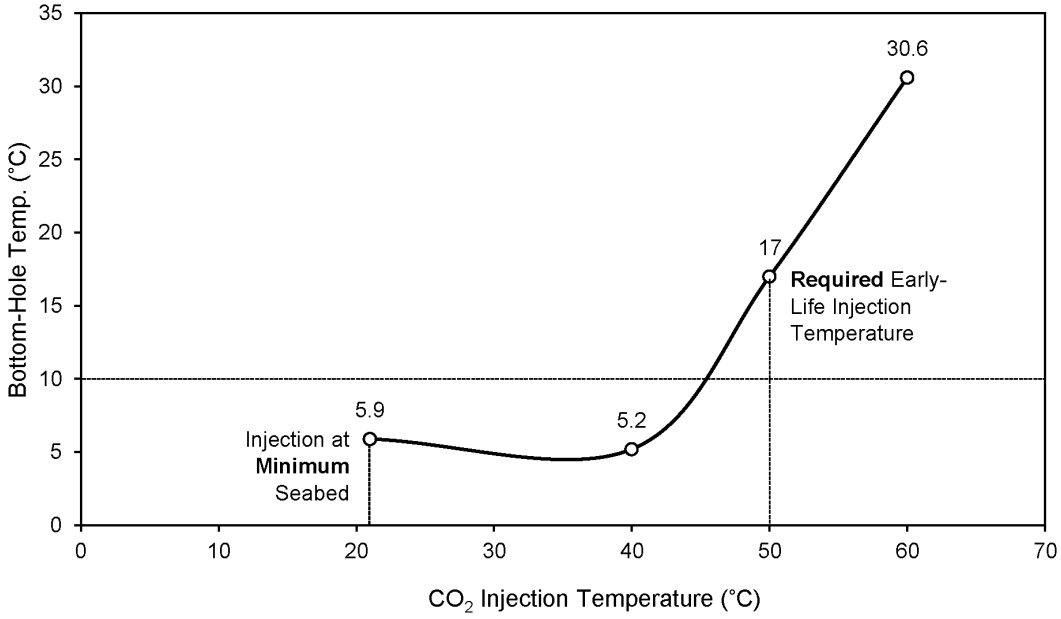


Figure 5—Effects of CO_2 Injection Temperature on the Well Bottom-Hole Temperature for the Intermediate Reservoir

For reservoirs shallower than the *intermediate* reservoir, a CO_2 injection temperature of 50°C is insufficient as the well is not deep enough for heat transfer from the surrounding rock formation to compensate the temperature drop in the wellbore from fluid expansion. Instead of increasing the injection temperature, an alternative is to limit the maximum injection flow into the shallower reservoirs allowing CO_2 to be warmed up by the surrounding formation. From Figure 6, CO_2 injection into the *shallow* reservoir during early-life operation must not exceed 37.5% of the well's design capacity to keep the bottom-hole temperature above 10°C. At these conditions, the maximum flow rate for the *deep* reservoir is also hydraulically restricted at 80% of the well's design capacity by the available injection pressure of 80 barg. Further raising the injection pressure beyond 80 barg will considerably impact the bottom-hole temperatures in the other reservoirs.

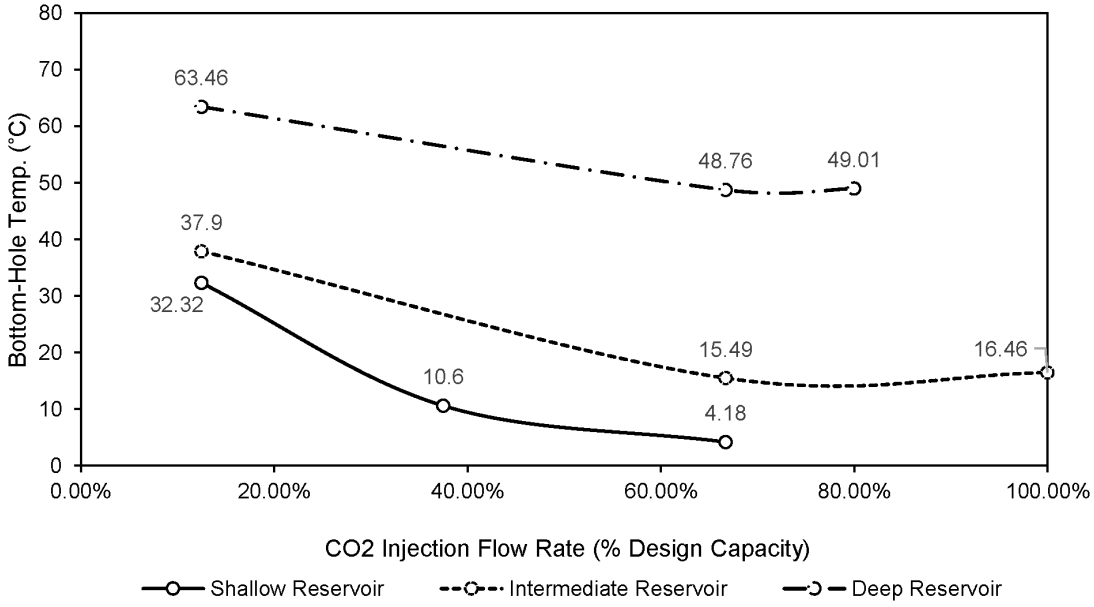


Figure 6—Early-Life Flow Rate Constraints for *Shallow*, *Intermediate*, and *Deep* Reservoirs

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Late-Life Operation

As the CO_2 fluid temperature at the well bottom-hole is a strong function of the reservoir pressure, another sensitivity analysis was performed to identify the minimum pressure of the depleted reservoir that will enable CO_2 to be injected at seabed temperature without the requirement for elevated operating temperatures. From Figure 7, all reservoirs at the CO_2 storage site must be pressurized above 50 barg to ensure that the CO_2 fluid temperature at the bottom-hole of each injection well remains above 10°C.

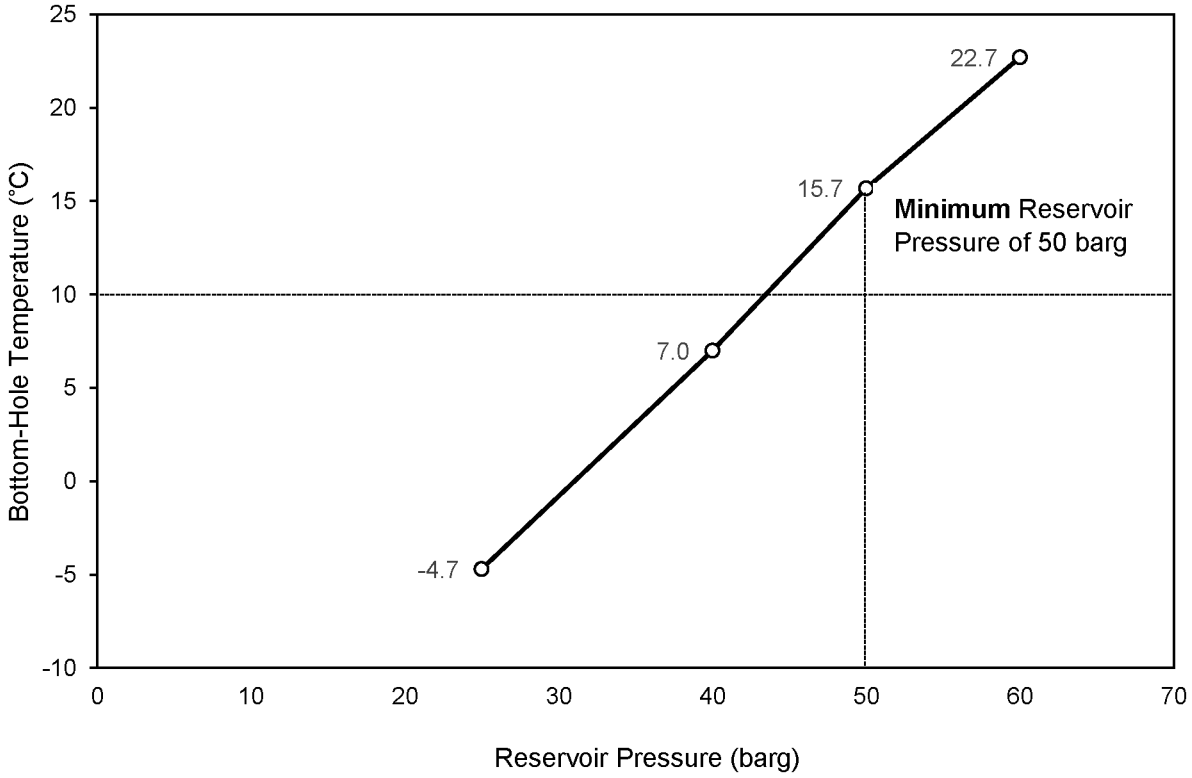


Figure 7—Effects of Reservoir Pressure on the Well Bottom-Hole Temperature

During late-life operation, CO_2 can be injected at seabed temperature as its behavior in the injection well is dominated by gravity / compression effects. The injection pressure, however, will have to be adjusted from 80 to 100 barg to account for the higher reservoir pressures.

Design of CO_2 Transport and Injection Network

Several design options were assessed to raise the CO_2 fluid temperature to 50°C upon arrival at the offshore injection platform. Additionally, the possibility to avoid low CO_2 fluid temperatures at the bottom-hole of each injection well by transporting and injecting CO_2 in gas phase was also studied. The three (3) options considered are described as follows:

- **Option 1: Pipeline Insulation**

  The CO_2 transport pipeline will be coated with dry insulation to preserve the CO_2 fluid temperature from the compression train at the gas processing plant. During the early-life reservoir pressurization phase, the discharge aftercooler of the final-stage CO_2 compressor will be partially bypassed enabling CO_2 to enter the transport pipeline at a maximum temperature of 110°C. Accounting for the system's turndown capacity, the thickness of the dry insulation is specified at

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a U-value of 1.5 W/m²K to ensure the CO_2 temperature arriving at the injection platform always remains above 50°C. A 14-inch CO_2 transport pipeline is required for this option.

- **Option 2: Offshore Electric Heater**

  A common electric heater will be provided on the offshore injection platform to raise the temperature of the CO_2 fluid stream from 20°C (minimum seabed temperature) at the pipeline outlet to 50°C upstream of the choke valves. Under this scenario, CO_2 will be transported in dense / supercritical phase through a 12-inch transport pipeline. Due to the large load of the electric heater, renewable power generation at the injection platform is not feasible. Instead, subsea power cables are necessary to supply electricity from the gas processing plant to the electric heater resulting in a relatively higher GHG footprint for this option.

- **Option 3: Gas-Phase CO_2 Injection**

  Gaseous CO_2 will be transported and injected into the depleted reservoir to reduce the J-T effect across the choke valves. In the early-life reservoir pressurization phase, the operating pressure at the injection platform will be restricted at 50 barg for gas-phase operation. With the lower density of CO_2 gas, a 24-inch transport pipeline is required in this option for the same system design capacity.

A techno-economic analysis was performed for the three (3) options listed above, and the comparative results are summarized in Table 4.

Table 4—Techno-Economic Comparison for Different CO_2 Transport Pipeline and Injection Platform Design Options

Design Option	Pipeline Insulation	Offshore Electric Heater	Gas-Phase CO_2 Injection
Pipeline Size	14-inch	12-inch	24-inch
Transport CO_2 Phase	Supercritical	Dense / Supercritical	Gas
Energy Consumption	Baseline	+50%	-10%
Pipeline Insulation	1.5 W/m²K	Typical Concrete Coating	Typical Concrete Coating
Relative CAPEX	Baseline	+40%	+10%
Relative OPEX	Baseline	+40%	-10%
GHG Footprint	Baseline	+60k tCO_2/y	Similar

Based on the techno-economic comparison in the table above, the option to insulate the CO_2 transport pipeline was selected as it requires the least capital upfront. When compared to installing an offshore electric heater, insulating the CO_2 transport pipeline is also more energy efficient as the heat of compression from the CO_2 compressors can be preserved resulting in no further GHG emission from implementing this option. For gas-phase CO_2 injection, a significantly larger pipeline (24-inch) is required leading to challenges in terms of operation and maintenance. More importantly, the total CO_2 injection flow during early-life operation cannot be sustained with gas-phase CO_2 injection without increasing the well count and/or the injection tubing size given that the upstream choke pressure is limited at 50 barg.

Transient Operations

Although the low temperature concerns in the depleted reservoirs can be managed during steady state injection, transient operations such as initial start-up and restarts present a different challenge as the CO_2 stream will not be at the required injection temperature (i.e., 50°C) during these scenarios. Dynamic flow assurance analysis was then conducted using the integrated pipeline-well OLGA model to develop the operating philosophies allowing the CO_2 transport and injection network to be started up and restarted safely within the system's design boundaries.

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Initial Start-Up

During initial start-up, the CO_2 injection wells will be completed with a liquid column. CO_2 injection is scheduled to commence after the CO_2 transport pipeline is pressurized to its normal operating pressure (i.e., 80 barg). Significant J-T cooling across the injection choke valve is expected as the equilibrium pressure and temperature above the liquid column (downstream of the choke valve) is close to atmospheric conditions (i.e., 0 barg and ambient temperature). Without backpressure at the wellhead, the CO_2 fluid temperature downstream of the choke valve can reach -64.6°C for a short period of time. In order to avoid inner wall metal temperatures falling below the design constraints, a relative fast well ramp-up rate need to be considered. Regardless of the ramp-up rate, CO_2 injection into the *shallow* and *intermediate* reservoirs can result in well bottom-hole temperatures below 10°C. In these cases, a hydrate inhibitor will be injected until the bottom-hole temperature exceeds 10°C.

Restarts

Low temperature concerns in the wellbore also need to be circumvented when the CO_2 transport and injection network is restarted during early-life operation. If the system is shut down for a short period of time before the fluid temperature in the CO_2 transport pipeline falls below 50°C, the CO_2 injection wells can be restarted immediately. From the pipeline shut-in temperature profiles in Figure 8, the "no touch time" of the system is about two (2) hours.

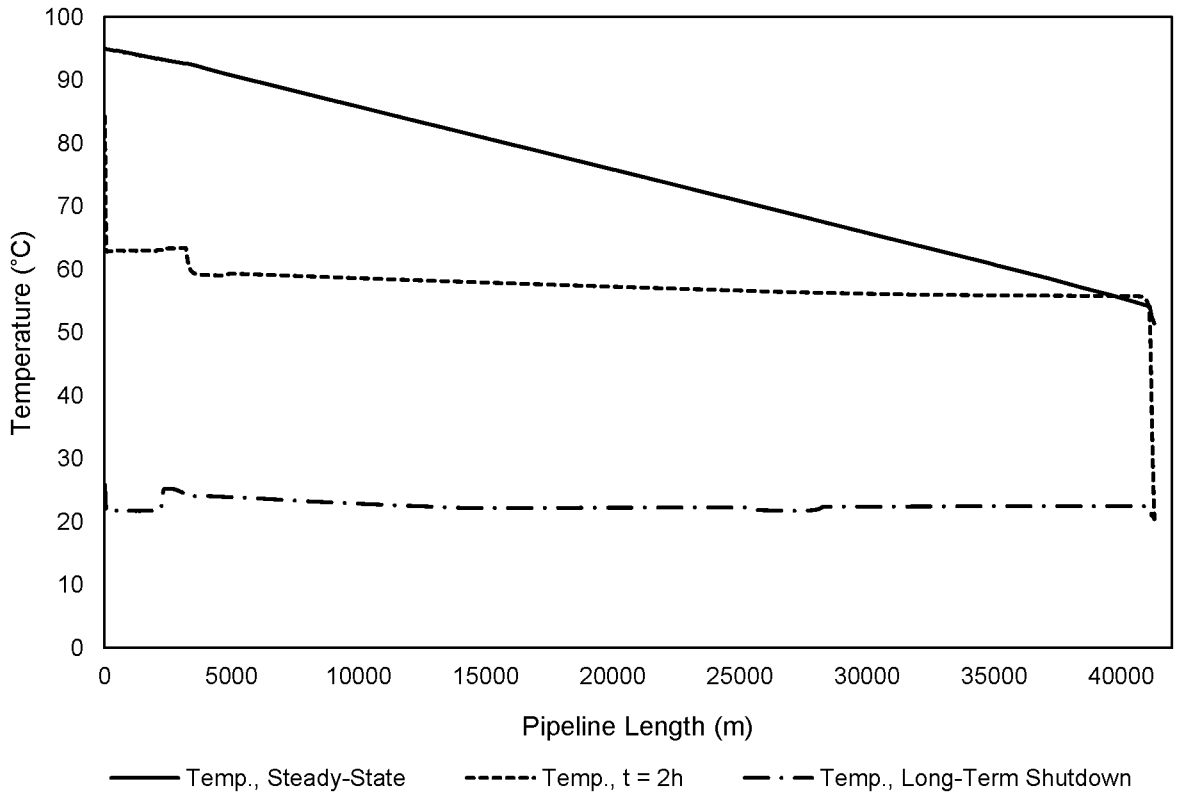


Figure 8—Temperature Profiles in the CO_2 Transport Pipeline During Shut-In Conditions

For prolonged shutdowns, the fluid temperature in the CO_2 transport pipeline will drop to the seabed temperature. To restart CO_2 injection in this scenario, it is necessary to start-up the system by first injecting CO_2 into the *deep* reservoir to bring the injection temperature back to 50°C (Figure 9). After stabilizing the pipeline outlet temperature, CO_2 injection into the other shallower reservoirs can then resume without any low temperature concerns even if the shut-in tubing head pressure is as low as 25 barg (Figure 10).

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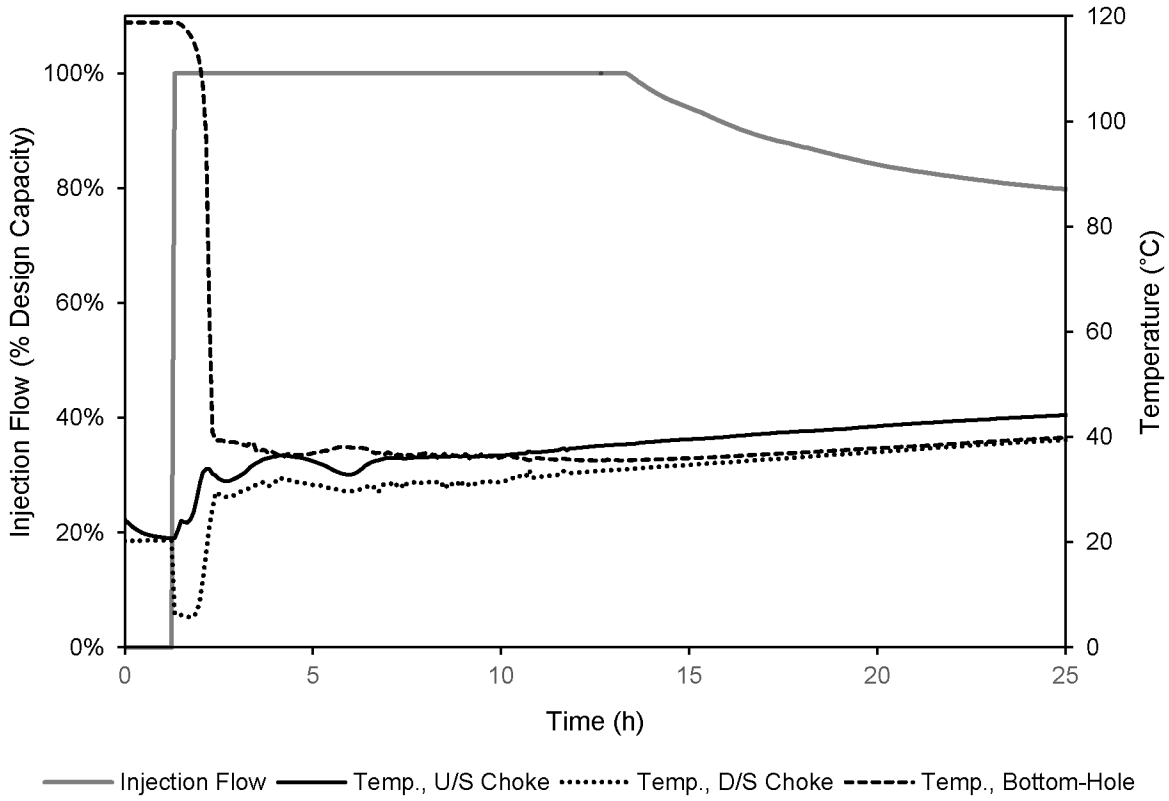


Figure 9—Flow Rate and Temperature Profiles for Restarting CO_2 Injection into the *Deep* Reservoir

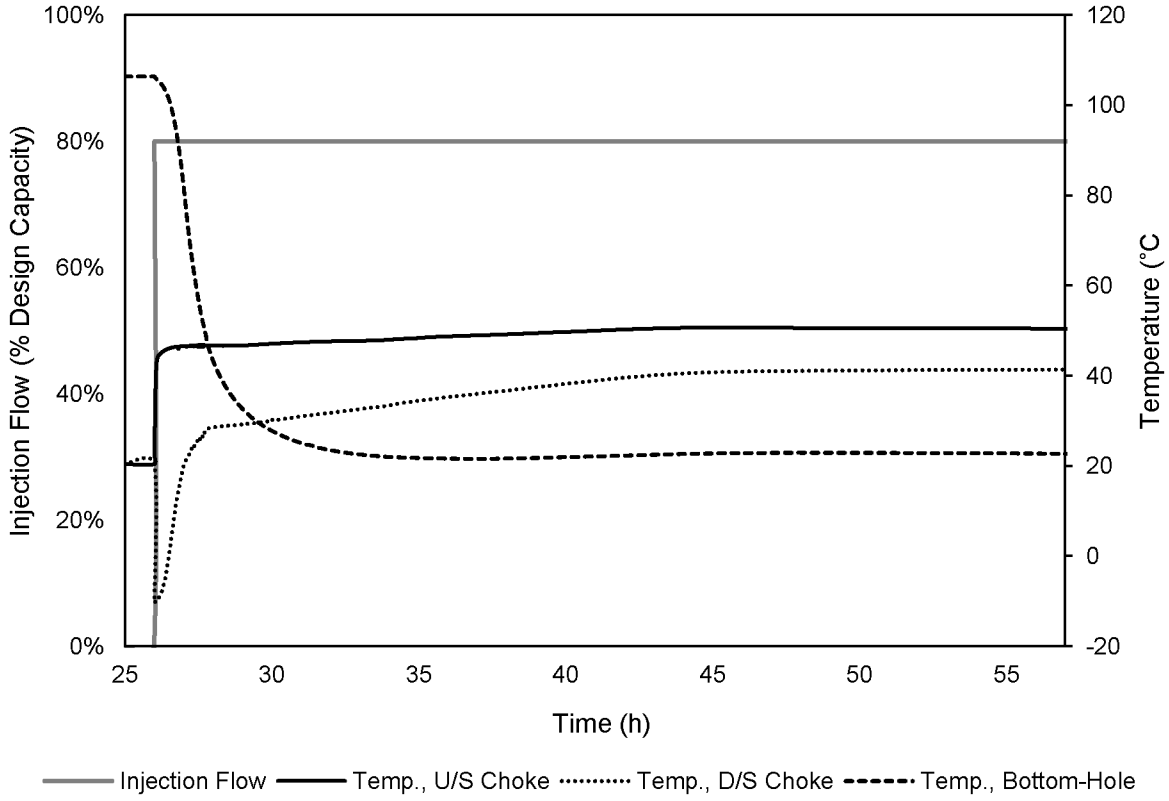


Figure 10—Flow Rate and Temperature Profiles for Restarting CO_2 Injection into the *Intermediate* Reservoir

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Conclusions and Recommendations

For this greenfield initiative, CO_2 injection in depleted hydrocarbon reservoirs proves to be not only viable but also sustainable in reducing GHG emission. However, low temperatures in the CO_2 injection wells – in particular at the Surface Control Subsurface Safety Valve (SCSSV) and bottom-hole are expected when the reservoir pressures of the CO_2 storages are relatively low. Understanding the CO_2 fluid behaviors and tendencies in the transport and injection network through fluid characterization exercises and steady state / dynamic flow assurance analysis is vital in defining the required pressure, temperature, and flow rate envelopes and optimizing facility design requirements to safely inject CO_2 into depleted reservoirs.

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